

line energy needs of all citizens, India needs to increase its primary energy supply by three to four times and its electricity generation capacity by about six times.

It is estimated that the total primary energy requirement in India would grow to 2,043 MTOE (Million Tonne of Oil Equivalent) by the year 2031-32. Even commercial energy requirement is expected to grow from an estimated 385 MTOE in 2006-07 to a whopping 1,858 MTOE by 2031-32. The Integrated Energy Policy has estimated that the requirement of power generation capacity would be almost 800 GW by the year 2031-32.

■ ROADBLOCKS ON THE WAY

The availability of reliable electricity supply has not kept pace with the burgeoning demand and increasing aspiration levels of a flourishing economy. So far, the power sector has been seen as a welfare activity. There is an urgent requirement to realise that it's an industrial and commercial venture which must provide for improvements and expansion, besides being an activity which is deemed to be a profitable venture for private investment.

Some major issues relating to this sector are:-

- Only 44 per cent rural households had access to electricity (2001 Census).
- Lack of commercial viability of State Electricity Boards/State Distribution Companies and modest private sector investment.
- Transmission and Distribution losses at 31 per cent being one of the highest in the world.
- Large scale theft of power. Amendment to the Electricity Act, 2003 for more stringent provisions has been approved by the Lok Sabha on 07/05/07. However, distribution being almost entirely with the States, enforcement of law is in the hands of the States.
- Distorted tariff structure with high industrial tariffs adversely affecting competitiveness of Indian industry combined with highly subsidized (in a few cases, even free supply of electricity) to agriculture leading to wasteful usage of water and ground water depletion.
- Operationalisation of open access in State level transmission and distribution is the key for operating an electricity market on competitive lines.
- Despite consistent generation growth of over 5% during the last four years, shortages remain alarmingly high due to the rapid increase in demand for electricity.
- Inefficient consumption of electricity requiring energy conservation and Demand Side Management (DSM).

■ RESURRECTING THE POWER SECTOR

The Ministry of Power has set a goal of 'Power for All by 2012'. This entails provision of adequate reliable power at affordable cost with access to all citizens. However, when one considers the bottlenecks mentioned in the last segment and the state of financial and operational health of the State utilities, the goal seemingly appears to be challenging.

However, the Ministry of Power has initiated a number of path-breaking initiatives to revamp the power sector to make it more vibrant in terms of planning, more people oriented in terms of implementation and more commercial in terms of day to day transactions. Various policies and programmes have been launched to run this sector on competitive lines with a view to bring the required generation capacities at an accelerated pace, to bring efficiency in generation segment; to ensure effective penetration of commercial energy in the rural areas in a time bound manner and to consolidate the electricity delivery system both in urban and rural areas.

In the recent past, series of legislative and policy initiatives have been taken with a view to ensure that every segment of electricity value chain i.e. generation, transmission and distribution develop on competitive lines in a more transparent framework. Pronouncement of new hydro policy on 3rd January, 2008 has been latest in the series. In order to encourage private producers to undertake hydro projects in difficult and remote areas, exemption from tariff based competitive bidding which is currently available to only public sector undertakings upto January, 2011, is now also available to private hydro projects. New hydro policy further, initiates that an additional 1% free power over and above 12% will be earmarked for a Local Area Development Fund aimed at providing a regular stream of revenue for income generation, infrastructure creation and welfare

schemes in the effected areas. With a view to provide a better package to the project effected families, policy further provides that each such family will get 100 units of electricity every month for a period of 10 years free of charge and project authorities will construct houses at resettled sites exclusively for project effected families.

A new policy paradigm for the development of power sector in the country has been created by the Electricity Act 2003 which has abolished the monopoly of the State Electricity Boards created through the Electricity (Supply) Act, 1948 and has evolved a new competitive framework for the development of the power sector in the country with focus on the consumer and safeguarding of his interests by independent Regulatory Commissions. Furthermore, it has eliminated / reduced entry barriers in the entire chain of the electricity supply business.

The response towards these initiatives from all the stakeholders has been overwhelming. Power sector is at centre stage now. All the stakeholders including private sector, lending sector and investors are no longer shy in investing in power sector and instead this sector is being looked as a most preferred destination for parking the investments. Participation of private sector in the generation segment has increased significantly. In the year 2006-07 projects aggregating over 6000 MW were ordered in the private sector. In addition during 2007-08 so far projects aggregating 9950 MW have been ordered in the private sector alone. Power sector IPOs floated in the recent past have also received overwhelming response. In 2007 IPOs of PFC and PGCIL were oversubscribed by over 77 times and 65 times respectively. Most recently, despite stock market being subdued, the IPO of REC was oversubscribed by near 28 times.

■ EYEING ACCELERATED POWER GENERATION

After the United States, China, Japan, Russia and Canada, India ranks sixth in terms of electricity generation. At the time of inception of planning era, generating capacity in the country was only 1750 MW which increased to 1,41,500 MW now. Growth in generation has been consistently above 5% in last few years. During the current financial year growth rate of 6.5% have been recorded up to February 2008.

The central sector's share in generation had gradually increased from 12 per cent in 1979 to 34 per cent on December 31, 2007. On the other hand, the share of state sector has declined from 82.5 per cent to 53 per cent during the same period. Even private sector's involvement has gone up from 5.2 per cent to 13 per cent.

The Eleventh Plan envisages a capacity addition programme of 78,577 MW. Out of this, 50 per cent would be contributed by the central sector. The remaining 50 per cent of capacity addition would be done either under the state sector or through private sector projects facilitated by the states. In the 1st year of the 11th Plan i.e. 2007-08 so far 8509 MW have been commissioned which is an all time record for a single year. Given the current status of 11th Plan projects aggregating over 60,000 MW are under construction. Thus, over 87% of capacity addition target is already at the implementation stage.

■ Development of Ultra Mega Power Projects

The development of Ultra Mega Power Projects (UMPP) has been identified as the thrust area. These large sized projects involve an estimated investment of about USD 4 billion. All these projects are being developed on a Build, Own and Operate (BOO) basis. The project developers have been identified on the basis of tariff based competitive bidding which is in line with one of the key objectives of the Electricity Act- 2003 — promotion of competition.

Ultra Mega Power Projects in Mundra (Gujarat), Sasan (Madhya Pradesh) and Krishnapatnam (Andhra Pradesh) are already underway after being handed over to successful bidders — Tata Power Company Limited in respect of Munda UMPP and M/s Reliance Power Limited in respect of Sasan and Krishnapatnam UMPPs. Furthermore, bidding process in respect of Talaiya (Jharkhand) has been initiated. At Request for Qualification (RFQ) stage 13 bids were received. These projects will meet the power needs of many states and distribution companies. The consultation process with other state governments is currently on to identify the feasible sites to further the development of these mega projects.

At the time of launching the UMPP initiative, there was general belief that tariff arrived at through this process would be less in comparison to the tariff being